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Backup power supply with charge control

Abstract

A power supply device includes a main battery, a backup battery, a first charge circuit, a second charge circuit, a first transistor, a second transistor, a switch element, and a control circuit. The first transistor has a control terminal for receiving a first control voltage, a first terminal coupled to the main battery, and a second terminal coupled to a first supply node. The second transistor has a control terminal for receiving a second control voltage, a first terminal coupled to the backup battery, and a second terminal coupled to a midway node. The switch element selectively couples the midway node to a second supply node according to a third control voltage. The control circuit monitors a first supply voltage at the first supply node, and generates the third control voltage according to the first supply voltage.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority of Taiwan Patent Application No. 113101337 filed on Jan. 12, 2024, the entirety of which is incorporated by reference herein.

BACKGROUND OF THE INVENTION

Field of the Invention

(2) The disclosure generally relates to a power supply device, and more specifically, to a power supply device with high output stability.

Description of the Related Art

(3) If the AC (Alternating Current) power source or the main battery of a conventional power

supply device is removed, an instant power shut-down event may occur, which may result in inconvenience for the user. Accordingly, there is a need to propose a novel solution for solving this problem of the prior art.

BRIEF SUMMARY OF THE INVENTION

(4) In an exemplary embodiment, the invention is directed to a power supply device for providing electric power to a system end. The power supply device includes a main battery, a backup battery, a first charge circuit, a second charge circuit, a first transistor, a second transistor, a switch element, and a control circuit. The first charge circuit is coupled to a first supply node, and is configured to generate a first control voltage. The first charge circuit selectively receives an AC (Alternating Current) voltage. The first transistor has a control terminal for receiving the first control voltage, a first terminal coupled to the main battery, and a second terminal coupled to the first supply node. The second charge circuit is coupled between a midway node and the first supply node, and is configured to generate a second control voltage. The second transistor has a control terminal for receiving the second control voltage, a first terminal coupled to the backup battery, and a second terminal coupled to the midway node. The switch element selectively couples the midway node to a second supply node according to a third control voltage. The control circuit monitors a first supply voltage at the first supply node. The control circuit also generates the third control voltage according to the first supply voltage.

(5) In some embodiments, if the first supply voltage is higher than or equal to the threshold voltage, the switch element will be opened. If the first supply voltage is lower than the threshold voltage, the switch element will be closed.

(6) In some embodiments, the first charge circuit is a buck converter.

(7) In some embodiments, the second charge circuit is a buck-boost converter.

(8) In some embodiments, each of the first transistor and the second transistor is an NMOSFET (N-type Metal Oxide Semiconductor Field Effect Transistor).

(9) In some embodiments, the power supply device further includes a first diode. The first diode has an anode coupled to the first supply node, and a cathode coupled to the system end.

(10) In some embodiments, the power supply device further includes a second diode. The second diode has an anode coupled to the second supply node, and a cathode coupled to the system end.

(11) In some embodiments, if the first charge circuit receives the AC voltage, the system end will be supplied by the first supply voltage at the first supply node.

(12) In some embodiments, if the first charge circuit does not receive any AC voltage and the main battery is enabled or operating normally, the system end will be supplied by the first supply voltage at the first supply node, and the first supply voltage will be generated by the main battery.

(13) In some embodiments, if the first charge circuit does not receive any AC voltage and the main battery is disabled or removed, the system end will be supplied by a second supply voltage at the second supply node, and the second supply voltage will be generated by the backup battery.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The invention can be more fully understood by reading the subsequent detailed description and examples with references made to the accompanying drawings, wherein:

(2) FIG. 1 is a diagram of a power supply device according to an embodiment of the invention; and

(3) FIG. 2 is a diagram of a power supply device according to another embodiment of the invention.

DETAILED DESCRIPTION OF THE INVENTION

(4) In order to illustrate the purposes, features and advantages of the invention, the embodiments and figures of the invention are shown in detail as follows.

(5) Certain terms are used throughout the description and following claims to refer to particular components. As one skilled in the art will appreciate, manufacturers may refer to a component by different names. This document does not intend to distinguish between components that differ in name but not function. In the following description and in the claims, the terms “include” and “comprise” are used in an open-ended fashion, and thus should be interpreted to mean “include, but not limited to . . .”. The term “substantially” means the value is within an acceptable error range. One skilled in the art can solve the technical problem within a predetermined error range and achieve the proposed technical performance. Also, the term “couple” is intended to mean either an indirect or direct electrical connection. Accordingly, if one device is coupled to another device, that connection may be through a direct electrical connection, or through an indirect electrical connection via other devices and connections.

(6) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(7) Furthermore, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(8) FIG. 1 is a diagram of a power supply device **100** according to an embodiment of the invention. As shown in FIG. 1, the power supply device **100** includes a main battery **110**, a backup battery **120**, a first charge circuit **130**, a second charge circuit **140**, a switch element **150**, a control circuit **160**, a first transistor M1, and a second transistor M2. It should be noted that the power supply device **100** may further include other components, such as a processor, a touch control module, a speaker, and/or a device housing, although they are not displayed in FIG. 1.

(9) The power supply device **100** can provide electric power to a system end **190**. For example, the system end **190** may be a tablet computer or a notebook computer, but it is not limited thereto. It should be understood that the system end **190** is an external device, which does not belong to any portion of the power supply device **100**.

(10) Both of the main battery **110** and the backup battery **120** store electric energy, and they are coupled to a ground voltage VSS. The types and styles of the main battery **110** and the backup battery **120** are not limited in the invention. In some embodiments, the main battery **110** has a relatively large battery capacity, and the backup battery **120** has a relatively small battery capacity.

(11) The first charge circuit **130** is coupled to a first supply node ND1, and is configured to generate a first control voltage VC1. It should be noted that the first charge circuit **130** can selectively receive an AC (Alternating Current) voltage VA. In some embodiments, the first charge circuit **130** is a buck converter. For example, the first charge circuit **130** may convert the AC voltage VA into a first supply voltage VD1 at the first supply node ND1. The DC (Direct Current) level of the first supply voltage VD1 may be lower than the RMS (Root Mean Square) level of the AC voltage VA.

(12) In some embodiments, the first transistor **M1** is an NMOSFET (N-type Metal Oxide Semiconductor Field Effect Transistor). Specifically, the first transistor **M1** has a control terminal (e.g., a gate) for receiving the first control voltage **VC1**, a first terminal (e.g., a source) coupled to the main battery **110**, and a second terminal (e.g., a drain) coupled to the first supply node **ND1**. In addition, a first body diode **DP1** may be built in the first transistor **M1**. The first body diode **DP1** has an anode coupled to the main battery **110**, and a cathode coupled to the first supply node **ND1**.

(13) In some embodiments, the first charge circuit **130** usually outputs the first control voltage **VC1** with a high logic level (or a logic “1”) for enabling the first transistor **M1**. In alternative embodiments, the first charge circuit **130** outputs the first control voltage **VC1** with a low logic level (or a logic “0”) for disabling the first transistor **M1**, and it is often used to prevent the main battery **110** from being overcharged.

(14) The second charge circuit **140** is coupled between a midway node **NE** and the first supply node **ND1**, and is configured to generate a second control voltage **VC2**. In some embodiments, the second charge circuit **140** is a buck-boost converter. For example, the second charge circuit **140** may convert the first supply voltage **VD1** into a midway voltage **VE** at the midway node **NE**. The DC level of the midway voltage **VE** may be higher or lower than the DC level of the first supply voltage **VD1**.

(15) In some embodiments, the second transistor **M2** is another NMOSFET. Specifically, the second transistor **M2** has a control terminal (e.g., a gate) for receiving the second control voltage **VC2**, a first terminal (e.g., a source) coupled to the backup battery **120**, and a second terminal (e.g., a drain) coupled to the midway node **NE**. In addition, a second body diode **DP2** may be built in the second transistor **M2**. The second body diode **DP2** has an anode coupled to the backup battery **120**, and a cathode coupled to the midway node **NE**.

(16) In some embodiments, the second charge circuit **140** usually outputs the second control voltage **VC2** with a high logic level for enabling the second transistor **M2**. In alternative embodiments, the second charge circuit **140** outputs the second control voltage **VC2** with a low logic level for disabling the second transistor **M2**, and it is often used to prevent the backup battery **120** from being overcharged.

(17) The switch element **150** can selectively couple the midway node **NE** to a second supply node **ND2** according to a third control voltage **VC3**. In some embodiments, if the third control voltage **VC3** has a high logic level, the switch element **150** will be closed, so as to couple the midway node **NE** to the second supply node **ND2**. At this time, a second supply voltage **VD2** at the second supply node **ND2** may be substantially equal to the midway voltage **VE**. In alternative embodiments, if the third control voltage **VC3** has a low logic level, the switch element **150** will be opened, so as to decouple the midway node **NE** from the second supply node **ND2**. At this time, the second supply voltage **VD2** at the second supply node **ND2** has nothing to do with the midway voltage **VE**.

(18) For example, the control circuit **160** may be implemented with an MCU (Microcontroller Unit), and it may include a detection module and a comparison module (not shown). The control circuit **160** can monitor the first supply voltage **VD1** at the first supply node **ND1**. The control circuit **160** can also generate the third control voltage **VC3** according to the first supply voltage **VD1**. Specifically, the control circuit **160** can compare the first supply voltage **VD1** with the threshold voltage **VTH**, and then determine the third control voltage **VC3** based on the comparison result. In some embodiments, if the first supply voltage **VD1** is higher than or equal to the threshold voltage **VTH**, the control circuit **160** will output the third control voltage **VC3** with a low logic level, such that the switch element **150** will be opened. In alternative embodiments, if the first supply voltage **VD1** is lower than the threshold voltage **VTH**, the control circuit **160** will output the third control voltage **VC3** with a high logic level, such that the switch element **150** will be closed. For example, the threshold voltage **VTH** may be about 5V, 7V or 8V, but it is not limited thereto.

(19) Generally, the power supply device **100** may be operated in either an AC mode, a main mode,

or a backup mode, and they will be described in the following embodiments.

(20) In the AC mode, if the first charge circuit **130** receives the AC voltage VA, the system end **190** will be supplied by the first supply voltage VD1 at the first supply node ND1. At this time, the first supply voltage VD1 is generated by the first charge circuit **130** according to the AC voltage VA, and the midway voltage VE is generated by the second charge circuit **140** according to the first supply voltage VD1. Furthermore, the main battery **110** can be charged by using the first supply voltage VD1, and the backup battery **120** can be charged by using the midway voltage VE. It should be noted that because the switch element **150** is opened in the AC mode, the second supply node ND2 of the power supply device **100** cannot be arranged to provide any electric power to the system end **190**.

(21) In the main mode, if the first charge circuit **130** does not receive any AC voltage VA and the main battery **110** is enabled or operating normally, the system end **190** will be supplied by the first supply voltage VD1 at the first supply node ND1. At this time, the first supply voltage VD1 is generated by the main battery **110**, and the midway voltage VE is generated by the second charge circuit **140** according to the first supply voltage VD1. Furthermore, the main battery **110** can perform a discharge operation to provide the first supply voltage VD1, and the backup battery **120** can be charged by using the midway voltage VE. It should be noted that because the switch element **150** is opened in the main mode, the second supply node ND2 of the power supply device **100** cannot be arranged to provide any electric power to the system end **190**.

(22) In the backup mode, if the first charge circuit **130** does not receive any AC voltage VA and the main battery **110** is disabled or removed, the system end **190** will be supplied by the second supply voltage VD2 at the second supply node ND2. It should be noted that since the first supply voltage VD1 at the first supply node ND1 fast drops, the switch element **150** is closed in the backup mode. The backup battery **120** can perform another discharge operation to provide the midway voltage VE. The second supply voltage VD2 may be substantially equal to the midway voltage VE. That is, the second supply voltage VD2 is generated by the backup battery **120**.

(23) With the design of the invention, the proposed power supply device **100** can support the function of hot swap. In other words, even if the main battery **110** is disabled or temporarily removed, the power supply device **100** can still provide electric power by using the backup battery **120**, and it does not need to use any AC power source. Also, a first auxiliary current path of the main battery **110** can be formed by the first body diode DP1 of the first transistor M1, and a second auxiliary current path of the backup battery **120** can be formed by the second body diode DP2 of the second transistor M2. In other words, since the power supply device **100** of the invention provides very high output stability, it can significantly reduce the probability of the system end **190** being unexpectedly shut down.

(24) FIG. 2 is a diagram of a power supply device **200** according to another embodiment of the invention. FIG. 2 is similar to FIG. 1. In the embodiment of FIG. 2, the power supply device **200** further includes a first diode D1 and a second diode D2. Specifically, the first diode D1 has an anode coupled to the first supply node ND1, and a cathode coupled to the system end **190**. The second diode D2 has an anode coupled to the second supply node ND2, and a cathode coupled to the system end **190**. According to practical measurements, the incorporation of the first diode D1 and the second diode D2 helps to effectively enhance the isolation between the first supply node ND1 and the second supply node ND2. In addition, the AC voltage VA can be provided by an AC power source **270**, which does not belong any portion of the power supply device **200**. For example, if the first charge circuit **130** is coupled to the AC power source **270**, the first charge circuit **130** may receive the AC voltage VA. Conversely, if the first charge circuit **130** is decoupled from the AC power source **270** or the AC power source **270** is removed, the first charge circuit **130** may not receive any AC voltage VA. Other features of the power supply device **200** of FIG. 2 are similar to those of the power supply device **100** of FIG. 1. Thus, the two embodiments can achieve similar levels of performance.

(25) The invention proposes a novel power supply device. According to practical measurements, the power supply device using the aforementioned design has at least the advantages of supporting the hot swap and enhancing the output stability, and therefore it is suitable for application in a variety of devices.

(26) Note that the above voltages, currents, resistances, inductances, capacitances and other element parameters are not limitations of the invention. A designer can adjust these settings according to different requirements. The power supply device of the invention is not limited to the configurations of FIGS. 1-2. The invention may merely include any one or more features of any one or more embodiments of FIGS. 1-2. In other words, not all of the features displayed in the figures should be implemented in the power supply device of the invention. Although the embodiments of the invention use MOSFET as examples, the invention is not limited thereto, and those skilled in the art may use other types of transistors, such as BJT (Bipolar Junction Transistor), JFET (Junction Gate Field Effect Transistor), FinFET (Fin Field Effect Transistor), etc., without affecting the performance of the invention.

(27) Use of ordinal terms such as “first”, “second”, “third”, etc., in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another or the temporal order in which acts of a method are performed, but are used merely as labels to distinguish one claim element having a certain name from another element having the same name (but for use of the ordinal term) to distinguish the claim elements.

(28) It will be apparent to those skilled in the art that various modifications and variations can be made in the invention. It is intended that the standard and examples be considered exemplary only, with the true scope of the disclosed embodiments being indicated by the following claims and their equivalents.

Claims

1. A power supply device for providing electric power to a system end, comprising: a main battery; a backup battery; a first charge circuit, coupled to a first supply node, and generating a first control voltage, wherein the first charge circuit selectively receives an AC (Alternating Current) voltage; a first transistor, wherein the first transistor has a control terminal for receiving the first control voltage, a first terminal coupled to the main battery, and a second terminal coupled to the first supply node; a second charge circuit, coupled between a midway node and the first supply node, and generating a second control voltage; a second transistor, wherein the second transistor has a control terminal for receiving the second control voltage, a first terminal coupled to the backup battery, and a second terminal coupled to the midway node; a switch element, selectively coupling the midway node to a second supply node according to a third control voltage; and a control circuit, monitoring a first supply voltage at the first supply node, wherein the control circuit generates the third control voltage according to the first supply voltage; wherein each of the first transistor and the second transistor is an NMOSFET (N-type Metal Oxide Semiconductor Field Effect Transistor).

2. The power supply device as claimed in claim 1, wherein if the first supply voltage is higher than or equal to a threshold voltage, the switch element is opened, and if the first supply voltage is lower than the threshold voltage, the switch element is closed.

3. The power supply device as claimed in claim 1, wherein the first charge circuit is a buck converter.

4. The power supply device as claimed in claim 1, wherein the second charge circuit is a buck-boost converter.

5. The power supply device as claimed in claim 1, further comprising: a first diode, wherein the first diode has an anode coupled to the first supply node, and a cathode coupled to the system end.

6. The power supply device as claimed in claim 5, further comprising: a second diode, wherein the

second diode has an anode coupled to the second supply node, and a cathode coupled to the system end.

7. The power supply device as claimed in claim 1, wherein when the first charge circuit receives the AC voltage, the system end is supplied by the first supply voltage at the first supply node.

8. The power supply device as claimed in claim 1, wherein when the first charge circuit does not receive any AC voltage and the main battery is enabled or operating normally, the system end is supplied by the first supply voltage at the first supply node, and the first supply voltage is generated by the main battery.

9. The power supply device as claimed in claim 1, wherein when the first charge circuit does not receive any AC voltage and the main battery is disabled or removed, the system end is supplied by a second supply voltage at the second supply node, and the second supply voltage is generated by the backup battery.
